Experiment 6: POS Tagging and Named Entity Recognition (NER)

A. POS Tagging (Parts of Speech Tagging)

```
import nltk

nltk.download("punkt", quiet=True)
nltk.download("punkt_tab", quiet=True)
nltk.download("averaged_perceptron_tagger_eng", quiet=True)
```

```
True
```

```
import nltk
from nltk.tokenize import word_tokenize

text = "Natural Language Processing enables computers to understand language

tokens = word_tokenize(text)
pos_tags = nltk.pos_tag(tokens)

print("Input Text:", text)
print("\nWord and POS Tags:")
print(pos_tags)

print("\nIndividual Tags:")
for word, tag in pos_tags:
    print(f"{word:15} -> {tag}")
```

```
Input Text: Natural Language Processing enables computers to understand langu

Word and POS Tags:
[('Natural', 'JJ'), ('Language', 'NNP'), ('Processing', 'NNP'), ('enables', '

Individual Tags:
Natural         -> JJ
Language        -> NNP
Processing      -> NNP
enables         -> VBZ
computers       -> NNS
to              -> TO
understand      -> VB
language        -> NN
.               -> .
```

B. Named Entity Recognition (NER)

```
import nltk

nltk.download("punkt", quiet=True)
nltk.download("punkt_tab", quiet=True)
nltk.download("averaged_perceptron_tagger_eng", quiet=True)
nltk.download("maxent_ne_chunker_tab", quiet=True)
nltk.download("words", quiet=True)
```

```
True
```

```
import nltk
from nltk.tokenize import word_tokenize

text = "Elon Musk founded SpaceX in the United States."

tokens = word_tokenize(text)
pos_tags = nltk.pos_tag(tokens)

# Recognize named entities
named_entities = nltk.ne_chunk(pos_tags)

print("Input Text:", text)
print("\nNamed Entity Recognition Tree:")
print(named_entities)

print("\nExtracted Entities:")

for subtree in named_entities:
    if isinstance(subtree, nltk.Tree):
        entity_name = " ".join(
            word for word, tag in subtree.leaves()
        )
        entity_type = subtree.label()
        print(f"{entity_name} -> {entity_type}")
```

```
Input Text: Elon Musk founded SpaceX in the United States.

Named Entity Recognition Tree:
(S
  (PERSON Elon/NNP)
  (PERSON Musk/NNP)
  founded/VBD
  (ORGANIZATION SpaceX/NNP)
  in/IN
  the/DT
  (GPE United/NNP States/NNPS)
  ./.)

Extracted Entities:
Elon -> PERSON
Musk -> PERSON
```

```
SpaceX -> ORGANIZATION
United States -> GPE
```